SHL Assessment Recommendation Agent – Approach Document

1. Problem Statement

Developed an AI agent to recommend SHL assessments based on hiring requirements. The system supports multi-turn conversations, clarification handling, assessment comparison, and guardrails.

2. Architecture

FastAPI
↓
LangChain Agent
↓
Tools
(Retrieval | Clarification | Comparison | Guardrails)
↓
FAISS
↓
SHL Catalog JSON

Design principles:

- Modular architecture
- Separation of concerns
- Tool-calling agent design
- Stateless API

3. Design Choices

Component	Choice
API	FastAPI
LLM	Gemini 2.5 Flash
Framework	LangChain
Embeddings	all-MiniLM-L6-v2
Vector Store	FAISS
Data Source	SHL JSON Catalog
Retrieval	Semantic Search

4. Retrieval Setup

Assessment records were embedded using Sentence Transformers and indexed in FAISS for semantic retrieval.

Retrieval strategy:

- Top-5 similarity search
- Metadata preservation
- Semantic ranking
- No chunking required since each assessment is already a structured, self-contained document.

5. Context Engineering

Conversation history is passed to the agent to preserve context.

Approach:

- Context-aware recommendations
- Clarification-first workflow
- Catalog-grounded generation
- Recommendation only after sufficient information gathering
- Reduced hallucinations through retrieval grounding

6. Prompt Design

The prompt was designed to:

- Understand roles and experience levels

- Extract competencies
- Ask follow-up questions when needed
- Recommend at most five assessments
- Compare assessments when requested
- Reject salary, legal, and unrelated queries
- Recommend only catalog entries

7. Agent Tools

Retrieval Tool – Semantic assessment search
Clarification Tool – Identifies missing information
Comparison Tool – Compares assessments
Guardrail Tool – Rejects unsupported queries

8. Evaluation Approach

Example scenarios tested:

Query	Expected Result
Hiring Java developer	Clarification
Mid-level Java developer	Recommendations
Compare OPQ and GSA	Comparison
Salary question	Rejection

Metrics considered:

- Recommendation relevance
- Response quality
- Latency
- Manual evaluation

9. Challenges and Improvements

Challenges:

- SentenceTransformer environment issues
- StructuredTool errors
- Prompt length impacting quota usage
- Retrieval before clarification
- Tool invocation inconsistencies

Improvements:

- Reduced prompt size
- Retrieval-tool separation
- Improved recommendation formatting
- Clarification workflow
- Better JSON responses

10. Future Work

Potential enhancements:

- Hybrid retrieval
- Metadata filtering
- Re-ranking
- Session persistence
- Conversation memory
- Automated evaluation

11. AI Assistance Disclosure

AI-assisted tools were used for debugging, prompt refinement, architecture exploration, and implementation support. System design, integration, testing, and evaluation were performed manually.